MILITARY SPECIFICATION SHEET

ELECTRON TUBES, MAGNETRON

TYPES 4J57 THROUGH 4J59

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Pulsed, fixed frequency, integral magnet, rated peak power output 180 kw

ABSOLUTE RATINGS:

Parameter:	Ef	tpc	Du	epy	ib	pi	Pi	T (anode)	T (cathode stem)	Alt
Unit:	V	μs	---	kv	a	kw	W	°C	°C	ft
Maximum:	13.86	2.50	0.001	25	35	600	400	100	340	10,000
Minimum:	---	---	---	---	---	---	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mounting position: Any
Cathode: Unipotential, oxide coated

TEST CONDITIONS:

Parameter:	Ef	tk	tpc	Du	Ib	VSWR
Unit:	V	sec	μs	---	mAdc	---
TEST 1:						
Maximum:	---	120	---	---	---	---
:	12.6	---	---	---	---	---
Minimum:	---	---	---	---	---	---
TEST 2:						
Maximum:	---	---	1.10	---	---	1.05
:	6.0	---	---	0.001	30	---
Minimum:	---	---	0.90	---	---	---

Frequency		
Type	MHz	±
4J57	6,525	50
4J58	6,425	50
4J59	6,325	50

GENERAL:

Qualification - Required

Preheating - Test condition 1

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	3	---					
1301	Heater current	---	1		If	3.25	4.00	A
4306	Pulse voltage	---	2		epy	16	19	kv
4250	Power output	---	2	t = 300 sec (max)	Po	180	---	W
---	Fixed frequency	---	2	T (anode) = 90° C (approx)				
				4J57	F	6,475	6,575	MHz
				4J58	F	6,375	6,475	MHz
				4J59	F	6,275	6,375	MHz
4308	RF bandwidth	---	2		BW	---	3.0/tpc	MHz
4310	Frequency pulling factor	2	2	VSWR = 1.75:1	ΔF	---	15	MHz
	Quality conformance inspection, part 3							
---	Life-test provisions	---	2	Group D	t	500	---	hrs
---	Life-test end point							
4250	Power output	---	2		Po	150	---	W
	Periodic-check tests	1	---					
---	Cathode stem temperature	x	---	Ef = 12.6 V; t = 15 minutes	T	250	340	°C
4004	Mean temperature coefficient of frequency	---	2		ΔF/ΔT	---	0.25	MHz/°C
1047	Low-temperature operation	---	2	tk = 120 sec	---	---	---	---

NOTES:

1. If these tests have previously been performed by the tube manufacturer, and if the materiel, manufacturing process, and test requirements are the same as when the tests were performed, the subject test(s) need not be performed.
2. The tube shall remain stable and free from moding between 20 and 32 mAdc, when a 1.75 VSWR with variable phase is introduced in the transmission line.
3. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level of II.

Custodians:
Army - EL
Navy - EC
Air Force - 85

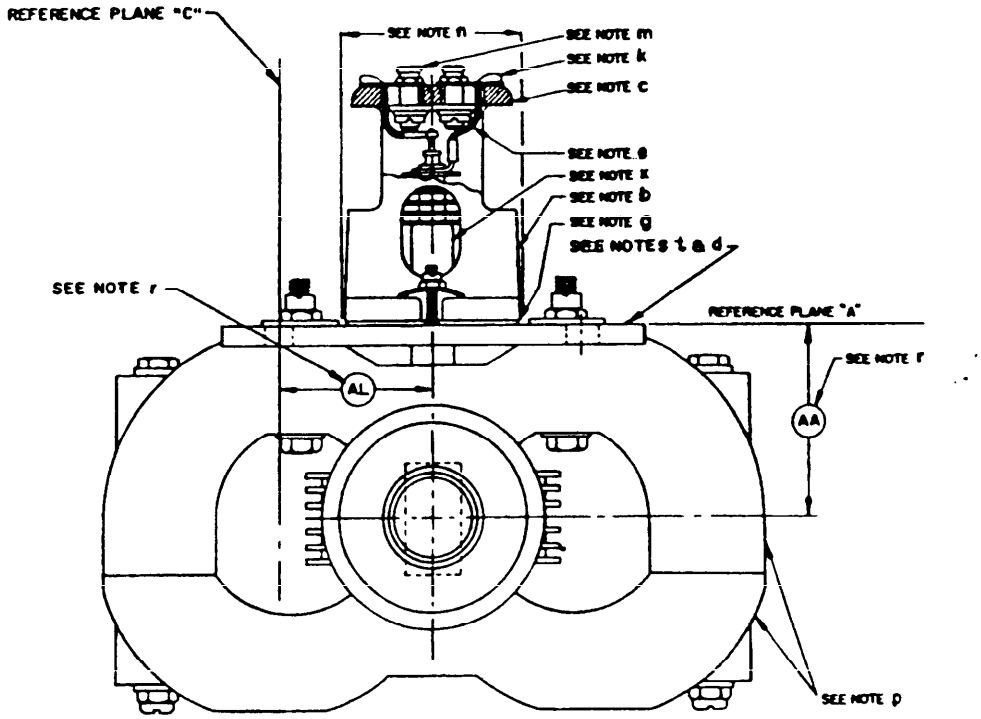
Preparing activity:
Navy - EC

(Project 5960-2374)

Review activities:
Army - EL
Navy -
Air Force - 11, 70, 85
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 19

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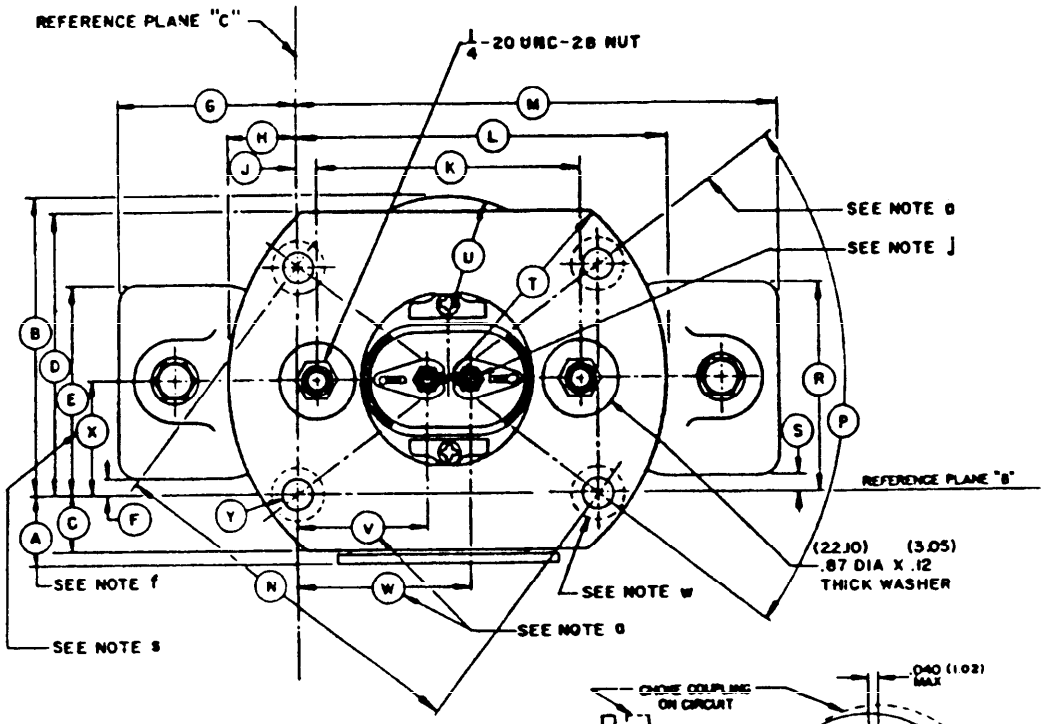


FRONT VIEW

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
AA	2.258 (57.35)	2.358 (59.89)
AL	1.743 (44.27)	1.803 (45.80)

FIGURE 1. Outline drawing of electron tube types 4J57 through 4J59.

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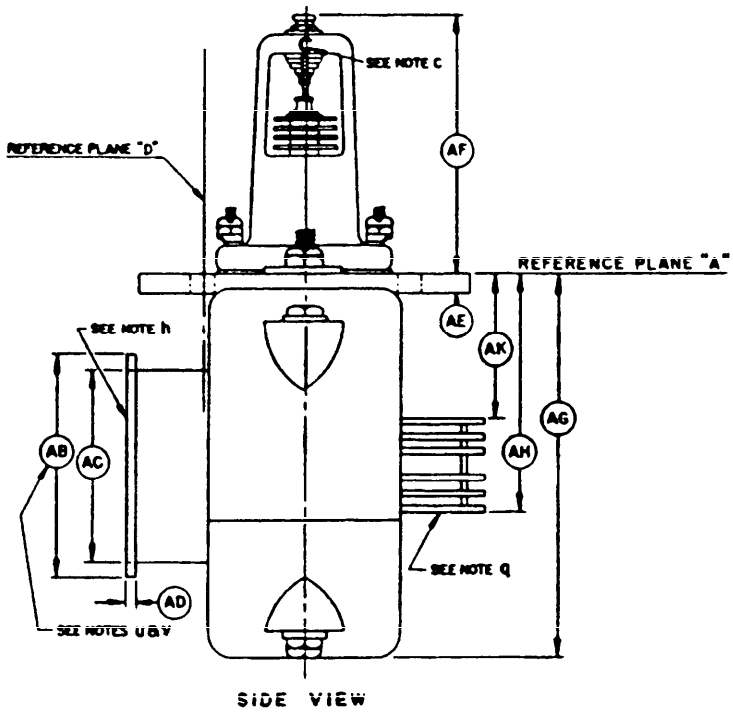


Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	.737 (18.72)	.797 (20.24)
C	.587 (14.91)	.639 (16.23)
D	3.356 (85.24)	3.408 (85.56)
H	.823 (20.90)	.875 (22.23)
L	4.370 (111.00)	4.422 (112.32)
N	4.490 (124.05)	4.510 (114.55)
P	75°35'	76°25'
Y	.370 (9.40)	.380 (9.65)
Quality conformance inspection, part 2		
B		3.182 (80.82)
E		2.720 (69.09)
F	.050 (1.27)	
G		2.563 (65.10)
M		6.187 (157.15)
R		2.720 (69.09)
S	.050 (1.27)	

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses
Quality conformance inspection part 3 (periodic check)	
	Nominal
J	.220 (5.59)
K	3.125 (79.39)
T	2.625 (66.68) R
U	1.500 (38.10) R
V	1.523 (38.68)
W	2.023 (51.38)
X	1.385 (35.18)

FIGURE 1. Outline drawing of electron tube types 4J57 through 4J59 - Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
AB	2.610 (66.29)	2.640 (67.06)
AC	2.357 (59.87)	2.387 (60.65)
AD	.110 (2.79)	.140 (3.56)
AE	.240 (6.10)	.260 (6.60)
AF	2.922 (74.22)	3.047 (77.39)
Quality conformance inspection, part 2		
AG		4.750 (120.65)
Quality conformance inspection, part 3 (periodic check)		
	Nominal	
AH	2.812 (71.42)	
AK	1.750 (44.45)	

FIGURE 1. Outline drawing of electron tube types 4J57 through 4J59 - Continued.

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NOTES:
a. The holes shall be within a radius of .023 (.58 mm) of the true location specified but shall be spaced .500 (12.70 mm) ±.010 (.25 mm) with respect to each other. The axis of holes shall be perpendicular to reference plane A within 3°.
b. Pyrex glass or approved equivalent.
c. Indicate common cathode connector by letter C.
d. All surfaces below top surface of mounting plate to be covered with painted finish except as specified by notes t and u.
e. Filament leads shall be flexible and slack.
f. Tolerance includes angular and lateral deviation of surface.
g. .03 (.79 mm) asbestos gasket (optional).
h. No portion of tube must extend outward beyond this surface.
j. Hex head banana pin locking jack .59 (15.09 mm) long hole .169 (4.29 mm) ±.005 (.13 mm).
k. Leads soldered to terminal clip.
m. Heater connection.
n. Any portion of the assembly extending above this surface shall be within a 1.109 (28.17 mm) radius of the true center of the plate except bolt heads and washers. Bolt heads and washers shall lie within a 1.250(31.75 mm) diameter circle centered as shown.
q. Radiator fins.
r. Applies to location of axis of output flange only.
s. Applies to location of centers of jack holes, bolts and washers only. See note a.
t. These surfaces to be free from paint.
u. Area within this diameter to be free from paint.
v. Outside diameter of output flange.
w. Minimum clearance between axis of plate holes and surface of magnet covering to be .312 (7.92 mm).
x. Measurement of cathode stem temperature is made by locating thermocouple on the kovar, adjacent to the kovar glass seal, on the cathode stem. Thermocouple is fastened with sauereisen cement or equivalent.
y. Alinement detail of tube to circuit coupling axis of choke coupling on circuit shall have a maximum displacement from the axis of output flange on tube of .040 (1.02 mm) in the plane of the electric field and .080 (2.03 mm) in the plane of the magnetic field. Axis of circuit coupling rectangular waveguide section shall be concentric with circuit coupling flange outer diameter within .010 (.25 mm).
z. If these dimensions have previously been inspected by the tube manufacturer and if the materiel, manufacturing process and dimensions are the same as when the inspections were performed, the subject inspection(s) need not be performed.

FIGURE 1. Outline drawing of electron tube types 4J57 through 4J59 - Continued.

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